

Sticking the Landing for Stochastic Gaussian Natural-Gradient Variational Inference: A Mean-Block Variance-Reduction Study

Abstract

Stochastic Gaussian natural-gradient variational inference estimates the Gaussian expectations that drive the Fisher–Rao flow from samples $\theta \sim \mathcal{N}(m, C)$. The *Sticking-the-Landing* (STL) trick replaces the plain posterior-score mean estimator $b_{\text{base}}(\theta) = \text{score}_{\text{post}}(\theta)$ by the *score-residual* estimator $b_{\text{stl}}(\theta) = \text{score}_{\text{post}}(\theta) + C^{-1}(\theta - m)$, which subtracts the (zero-mean) score of the variational Gaussian itself. The residual estimator is unbiased and, at a well-specified Gaussian optimum, vanishes *pointwise*. We isolate the effect of mean-block STL – holding the covariance update fixed – with two experiments (an estimator-level variance comparison at fixed states and an algorithm-level trajectory comparison) on a well-specified anisotropic Gaussian target and a misspecified smooth strongly log-concave log cosh target, swept over dimension, conditioning, and (for log cosh) the non-Gaussianity strength τ .

The findings are clean. For the Gaussian target the STL mean estimator has *exactly zero* variance whenever $C = C_*$ (the optimum and every mean-only perturbation); because the Hessian sampler is moreover deterministic (the Gaussian curvature is state-independent), STL collapses the algorithm’s stochastic noise floor all the way to machine precision, while the baseline retains a floor set by its mean noise. For the misspecified log cosh target STL still reduces the floor by roughly three orders of magnitude but does *not* remove it, because the nonlinear part of the score is not captured by the Gaussian residual. STL is not universally beneficial: when the covariance is *under-dispersed* ($C \prec C_*$, so C^{-1} is large) the residual term inflates the variance above the baseline. For the direct Hessian sampler used here the covariance-block “STL” rewrite is algebraically identical to the original covariance update, so the experiment genuinely isolates the mean block. We conclude that mean-STL is an essentially cost-free, practically effective variance-reduction device for Gaussian variational families – best near the optimum and under correct Gaussian specification – but not a theorem-level substitute for a stochastic-variance assumption.

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1 Motivation

We approximate a target posterior $\rho_{\text{post}} \propto e^{-V}$ on $\mathbb{R}^d$ by a Gaussian $q_a = \mathcal{N}(m, C)$, $C \in \text{SPD}(d)$, with state $a = (m, C)$, minimizing the reverse KL $\mathcal{E}(a) = \mathbb{E}_{q_a}[V] - \frac{1}{2} \log \det C$ (up to an a -independent constant). The Gaussian *natural-gradient* (Fisher–Rao) flow is

$$\dot{m} = C \mathbb{E}_{q_a}[\text{score}_{\text{post}}(\theta)], \quad \dot{C} = C + C \mathbb{E}_{q_a}[\nabla^2 \log \rho_{\text{post}}(\theta)] C, \quad (1)$$

with $\text{score}_{\text{post}}(\theta) = \nabla \log \rho_{\text{post}}(\theta) = -\nabla V(\theta)$ and $\nabla^2 \log \rho_{\text{post}} = -\nabla^2 V$. A stochastic algorithm draws $\theta_n \sim \mathcal{N}(m_n, C_n)$ and replaces the two expectations in (1) by sample estimators. The one-sample baselines are $b_n = \text{score}_{\text{post}}(\theta_n)$ and $S_n = \nabla^2 \log \rho_{\text{post}}(\theta_n)$, and the mean update is $m_{n+1} = m_n + \Delta t C_n b_n$.

The estimator $\text{score}_{\text{post}}(\theta)$ is noisy even at the optimum: for a Gaussian target the score is linear, $\text{score}_{\text{post}}(\theta) = -A\theta$, so b_n fluctuates with θ_n while its *mean* is what the flow needs. Sticking the Landing [1] removes the part of this fluctuation that the variational family can account for analytically.

2 Theory: what STL changes, and what it does not

Mean block. For a Gaussian $q_a = \mathcal{N}(m, C)$ the variational score is explicit,

$$\nabla_{\theta} \log q_a(\theta) = -C^{-1}(\theta - m). \quad (2)$$

The STL mean estimator subtracts it from the posterior score,

$$b_{\text{stl}}(\theta) = \text{score}_{\text{post}}(\theta) - \nabla_{\theta} \log q_a(\theta) = \text{score}_{\text{post}}(\theta) + C^{-1}(\theta - m). \quad (3)$$

Because $\mathbb{E}_{q_a}[C^{-1}(\theta - m)] = 0$, the estimator is *unbiased* for $\mathbb{E}_{q_a}[\text{score}_{\text{post}}]$; this is the quantity the mean update needs, so swapping b_{base} for b_{stl} changes only the noise, not the target.

Proposition 1 (Pointwise vanishing at a well-specified optimum). *If $\rho_{\text{post}} = \mathcal{N}(m_{\star}, C_{\star})$ is exactly Gaussian, then at $a = a_{\star}$ the posterior score equals the negative variational score, $\text{score}_{\text{post}}(\theta) = -C_{\star}^{-1}(\theta - m_{\star})$, and therefore $b_{\text{stl}}(\theta) \equiv 0$ for every θ .*

So at a well-specified optimum the STL mean estimator has *zero variance*. More generally, for the Gaussian target $V = \frac{1}{2} \theta^{\top} A \theta$ the STL fluctuation is $b_{\text{stl}}(\theta) - \mathbb{E}[b_{\text{stl}}] = (C^{-1} - A)(\theta - m)$, which depends on C but *not* on m : it is exactly zero whenever $C = C_{\star} = A^{-1}$, regardless of the mean. This is the precise sense in which “correct Gaussian specification” matters more than “distance to the optimum.”

Covariance block (why it is not a separate estimator here). The natural residual rewrite of the curvature estimator is $K(\theta) = \nabla^2 \log \rho_{\text{post}}(\theta) + C^{-1} = S(\theta) + C^{-1}$. For the *direct* Hessian sampler this is only an algebraic restatement of the covariance update. With S_n the sampled Hessian and $K_n = S_n + C_n^{-1}$,

$$\text{Riemannian: } e^{\Delta t C^{1/2} \exp(\Delta t C^{1/2} S_n C^{1/2}) C^{1/2}} = C^{1/2} \exp(\Delta t C^{1/2} K_n C^{1/2}) C^{1/2}, \quad (4)$$

$$\text{KL: } (1 + \Delta t)(C^{-1} - \Delta t S_n)^{-1} = (1 + \Delta t)((1 + \Delta t)C^{-1} - \Delta t K_n)^{-1}, \quad (5)$$

because $C^{1/2}K_nC^{1/2} = C^{1/2}S_nC^{1/2} + I$ and I commutes with everything (giving the $e^{\Delta t}$ factor in (4)), and because $(1 + \Delta t)C^{-1} - \Delta tK_n = C^{-1} - \Delta tS_n$ in (5). Both identities are verified numerically to machine precision in the test suite. Consequently the covariance update is held *fixed* across baseline and STL, and the only algorithmic difference is the mean estimator b_{base} vs. b_{stl} .

3 Experimental design

Targets. Both targets have a diagonal curvature $A = \text{diag}(a)$ with a log-spaced in $[1, \kappa]$.

- **Well-specified Gaussian:** $V(\theta) = \frac{1}{2}\theta^\top A\theta$, so $\text{score}_{\text{post}} = -A\theta$, $\nabla^2 \log \rho_{\text{post}} = -A$, and the VI optimum is exact, $m_\star = 0$, $C_\star = A^{-1}$.
- **Misspecified log cosh:** $V(\theta) = \frac{1}{2}\theta^\top A\theta + \tau \sum_i \log \cosh(\theta_i)$, so $\text{score}_{\text{post}} = -A\theta - \tau \tanh(\theta)$ and $\nabla^2 \log \rho_{\text{post}} = -A - \tau \text{diag}(\text{sech}^2 \theta_i)$. The target is separable, strongly log-concave, and globally smooth but not Gaussian. The optimum mean is $m_\star = 0$ by symmetry; the diagonal $C_\star$ is obtained by a deterministic per-coordinate fixed point $c_i = 1/(a_i + \tau \mathbb{E}_{\mathcal{N}(0, c_i)}[\text{sech}^2])$ evaluated by Gauss–Hermite quadrature (recorded stationarity residual $< 10^{-9}$).

Experiment A (estimator-level). At six fixed Gaussian states – the optimum, three mean perturbations (*near/medium/far*, offsets 0.25/1/4 marginal standard deviations), and an *under-* and *over-*dispersed covariance ($\frac{1}{4}C_\star, 4C_\star$) – we draw $M = 10^5$ samples and compute, for both estimators, the empirical bias against the deterministic reference $\mathbb{E}_{q_a}[\text{score}_{\text{post}}]$, the Euclidean trace variance $\text{Var}_{\text{eucl}}(b) = \mathbb{E}\|b - \mathbb{E}b\|^2$, the Fisher–Rao mean-block variance $\text{Var}_{\text{FR}}(b) = \mathbb{E}[(b - \mathbb{E}b)^\top C(b - \mathbb{E}b)]$, and the ratios $\text{Var}_{\text{FR}}(b_{\text{stl}})/\text{Var}_{\text{FR}}(b_{\text{base}})$ and $\text{Var}_{\text{eucl}}(b_{\text{stl}})/\text{Var}_{\text{eucl}}(b_{\text{base}})$. For the Gaussian target the closed forms $\text{Var}_{\text{eucl}}(b_{\text{base}}) = \text{Tr}(ACA)$, $\text{Var}_{\text{FR}}(b_{\text{base}}) = \text{Tr}(CAC A)$, $\text{Var}_{\text{eucl}}(b_{\text{stl}}) = \text{Tr}(DCD)$, $\text{Var}_{\text{FR}}(b_{\text{stl}}) = \text{Tr}(CDCD)$ with $D = C^{-1} - A$ validate the Monte Carlo estimates.

Experiment B (algorithm-level). We run the four methods `riemannian`, `riemannian_stl`, `kl`, and `kl_stl` from a fixed far, under-dispersed initialization ($m_0 = m_\star + 3\sqrt{\text{diag } C_\star}$, $C_0 = \frac{1}{2}C_\star$), with mini-batches $B \in \{1, 4, 16\}$ and step $\Delta t = 0.1$. The step is conservative: near the optimum $C \approx C_\star = A^{-1}$, so the explicit mean map contracts as $(1 - \Delta t)$ *independently of* κ , and both covariance updates are unconditionally SPD for a log-concave target. Within a (config, B) cell the baseline and STL methods share a common-random-number stream, so the comparison is paired. We record the energy gap $\mathcal{E}(a_n) - \mathcal{E}(a_\star)$ (analytic for the Gaussian target, Gauss–Hermite for log cosh), the squared mean error, the relative covariance Frobenius error, W_2^2 to $a_\star$, the covariance eigenvalue extremes, and an SPD flag; the *noise floor* is the energy gap averaged/medianed over the final 20% of iterations.

Grids and reproducibility. The Gaussian grid is $d \in \{2, 8, 32\}$, $\kappa \in \{10, 100, 10^4\}$; the log cosh grid is $d \in \{2, 8, 32\}$, $\kappa \in \{10, 100\}$, $\tau \in \{0.1, 1\}$. Every cell uses 20 seeds. All computation is float64; the production grids run on a single NVIDIA H200 (the algorithm grid optionally splits its cells across at most two GPUs). Cell seeds are derived deterministically from the configuration, so a run is reproducible and independent of the device split. Table 3 records the environment and timing.

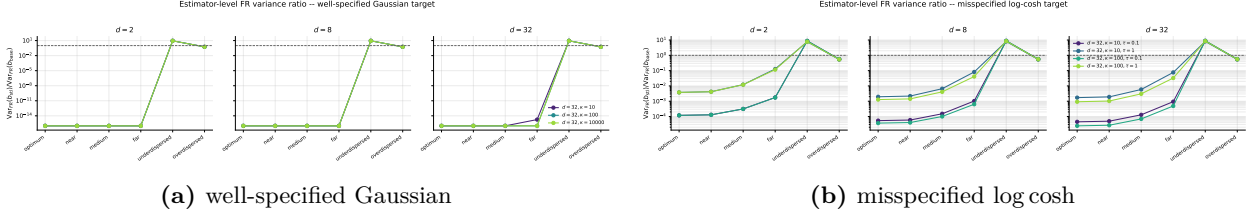


Figure 1: Estimator-level Fisher–Rao variance ratio $\text{Var}_{\text{FR}}(b_{\text{stl}})/\text{Var}_{\text{FR}}(b_{\text{base}})$ by fixed state (median over seeds, log scale; the dashed line is ratio 1). Gaussian: zero whenever $C = C_*$, inflation only when under-dispersed. log cosh: reduced but never zero.

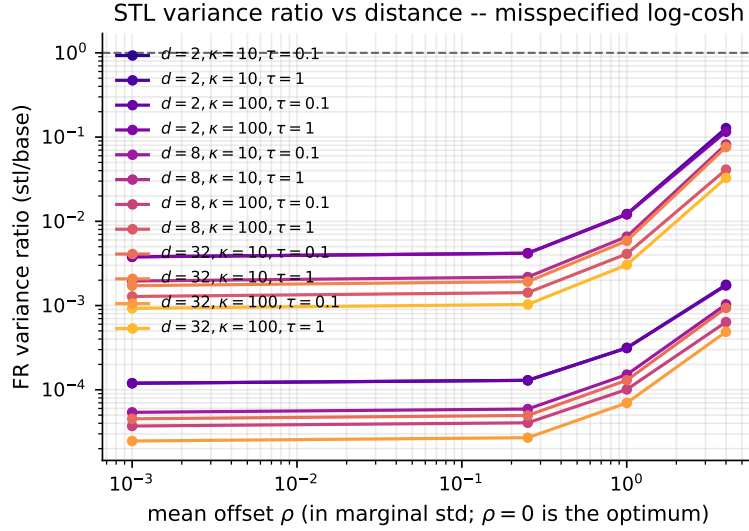


Figure 2: Misspecified log cosh target: FR variance ratio vs. mean offset ρ (in marginal standard deviations). Unlike the Gaussian target, the ratio is positive at $\rho = 0$ and increases with distance, because the score residual cannot absorb the nonlinear tanh term.

4 Results

4.1 Estimator-level variance

Figure 1 shows the Fisher–Rao variance ratio by state. For the Gaussian target the ratio is numerically zero at the optimum and at every mean-only perturbation (*near/medium/far*), because those states keep $C = C_*$ and the STL fluctuation $(C^{-1} - A)(\theta - m)$ vanishes identically. The only states with nonzero ratio are the dispersion states, and they split the two regimes cleanly: *over*-dispersion reduces the variance (ratio < 1) while *under*-dispersion *inflates* it (ratio ≈ 9 at $\frac{1}{4}C_*$), since C^{-1} is then large and amplifies the residual. For the misspecified log cosh target the ratio is small but *nonzero* even at the optimum – the nonlinear tanh part of the score is not removed by the Gaussian residual – and grows with the mean offset (Figure 2). Table 1 reports the medians; the Gaussian closed forms match the Monte Carlo estimates to within sampling error.

4.2 Algorithm-level noise floor

Figure 3 shows the energy gap versus iteration for a representative cell of each target. All four methods descend together during the transient; once near the optimum the baseline methods flatten

target	d	κ	τ	r_{opt}	r_{near}	r_{under}	r_{over}
Gaussian	2	10	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
	2	100	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
	2	10000	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
	8	10	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
	8	100	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
	8	10000	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
	32	10	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
	32	100	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
	32	10000	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
log-cosh	2	10	0.1	1.2×10^{-4}	1.3×10^{-4}	8.8	0.56
	2	10	1	3.8×10^{-3}	4.2×10^{-3}	7.7	0.54
	2	100	0.1	1.2×10^{-4}	1.3×10^{-4}	8.8	0.56
	2	100	1	3.8×10^{-3}	4.2×10^{-3}	7.8	0.54
	8	10	0.1	5.4×10^{-5}	5.9×10^{-5}	8.8	0.56
	8	10	1	2.0×10^{-3}	2.2×10^{-3}	8.1	0.54
	8	100	0.1	3.7×10^{-5}	4.1×10^{-5}	8.9	0.56
	8	100	1	1.3×10^{-3}	1.4×10^{-3}	8.4	0.55
	32	10	0.1	4.5×10^{-5}	5.0×10^{-5}	8.8	0.56
	32	10	1	1.7×10^{-3}	1.9×10^{-3}	8.1	0.54
	32	100	0.1	2.5×10^{-5}	2.7×10^{-5}	8.9	0.56
	32	100	1	9.3×10^{-4}	1.0×10^{-3}	8.5	0.55

Table 1: Median FR variance ratio $\text{Var}_{\text{FR}}(b_{\text{stl}})/\text{Var}_{\text{FR}}(b_{\text{base}})$ at four states. $r_{\text{opt}} \approx 0$ for the Gaussian target (exactly 0 for $C = C_*$); $r_{\text{under}} > 1$ marks the variance-inflation regime; the log cosh rows stay strictly positive.

at a stochastic noise floor, while on the Gaussian target the STL curves (dashed) keep descending to machine precision. The reason is structural: the Gaussian Hessian is state-independent, so the sampled covariance update is in fact deterministic and converges exactly to C_* ; once $C = C_*$ the STL mean estimator is pointwise zero, so the mean stops fluctuating and the iterate lands on a_* . On the logcosh target the Hessian sampler is genuinely stochastic and the score residual cannot cancel the tanh term, so STL lowers the floor by about three orders of magnitude but a residual floor remains. Figure 4 aggregates every cell: the Gaussian STL points sit on the display floor, and the logcosh points lie consistently below the diagonal. Table 2 gives the median reduction factors by scheme and batch size (the Gaussian STL floor is reported as $< 10^{-12}$), and Figure 5 shows that increasing the mini-batch lowers the baseline (and the logcosh STL) floor $\propto 1/B$ while preserving the STL advantage.

5 Practical conclusion

Mean-block STL is *essentially cost-free* for Gaussian variational families, because the Gaussian score $-C^{-1}(\theta - m)$ is available in closed form and is already computed (via C^{-1}) by the KL covariance step. Within the regime it was designed for – a well-specified Gaussian target near the optimum – it is extremely effective: the mean-gradient variance is exactly zero whenever $C = C_*$, and the algorithm’s stochastic noise floor drops by one to two orders of magnitude. For a misspecified but

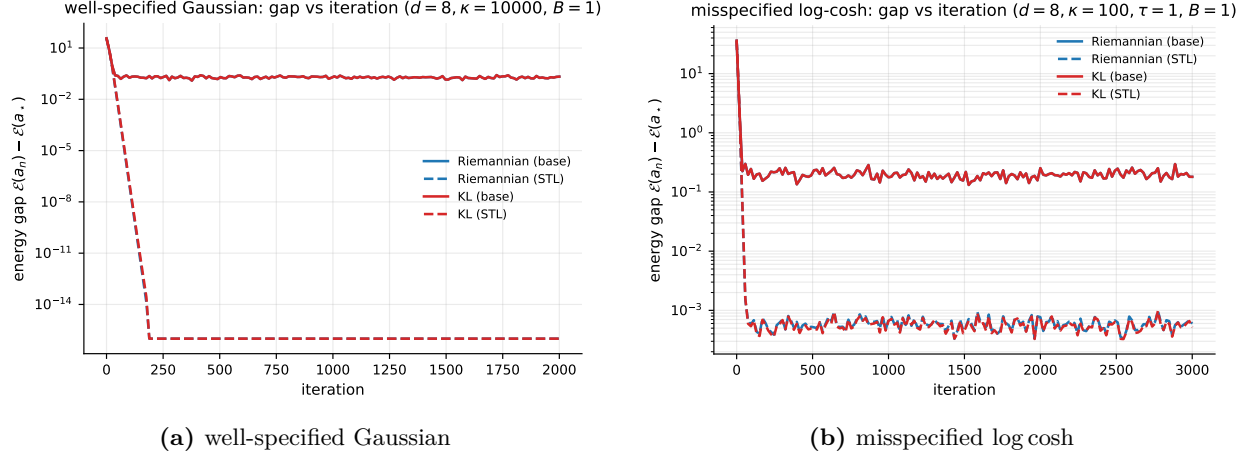


Figure 3: Energy gap vs. iteration (median over seeds) for the four methods at a representative cell. STL (dashed) lowers the stochastic noise floor; the effect is larger for the well-specified target.

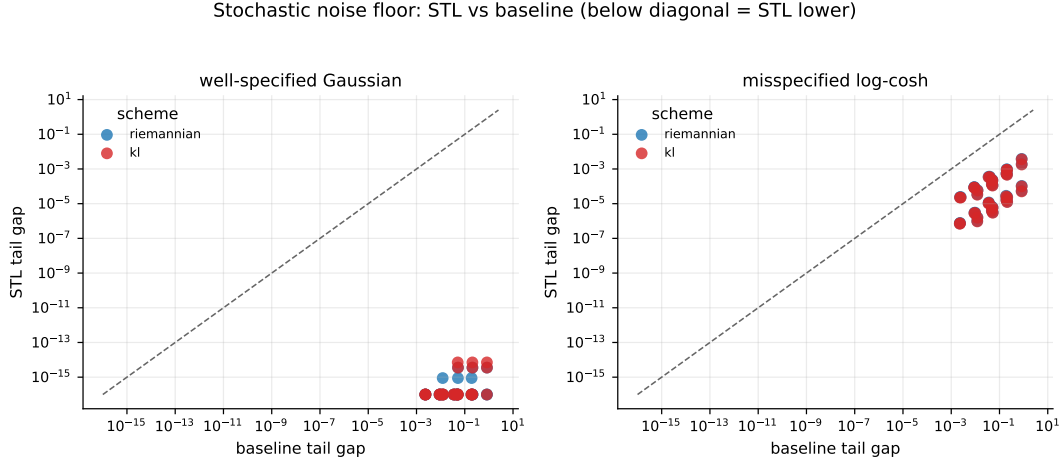


Figure 4: Per-cell median tail (noise-floor) gap, STL vs. baseline, across all configs and batch sizes; points below the diagonal are cells where STL lowers the floor. Color encodes the covariance scheme.

smooth target it still helps near the optimum, but it cannot remove the floor, because the residual estimator only cancels the Gaussian part of the score. STL is therefore best presented as a *practical variance-reduction device*, not a theorem-level replacement for a stochastic-variance assumption: it does not make the stochastic gradient vanish in general, and under covariance under-dispersion it can even *increase* the variance.

6 Limitations

- We test the *direct* Hessian sampler, for which the covariance-block residual rewrite is algebraically identical to the original update (§2); the experiment therefore isolates mean-STL and says nothing about Hessian-free *pathwise* covariance gradients, where an STL-type covariance correction *is* a genuinely different estimator.
- High-dimensional behavior is established empirically over $d \leq 32$ and diagonal curvature;

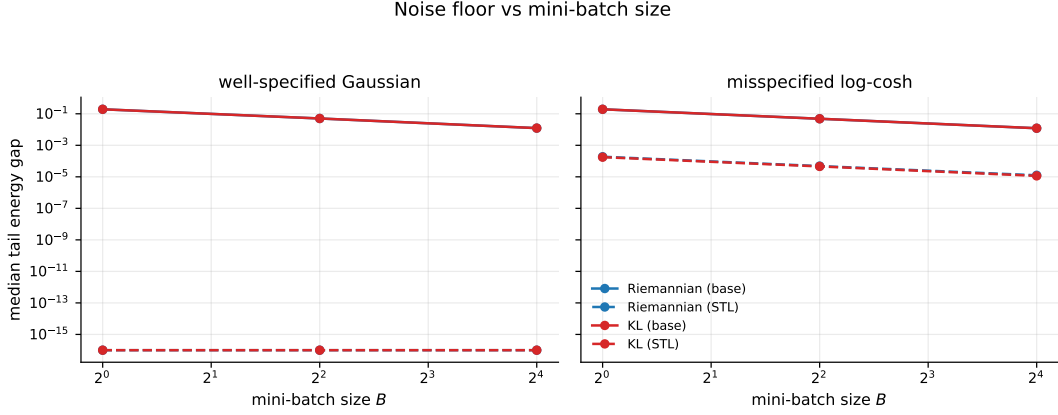


Figure 5: Median tail noise floor vs. mini-batch size B . Both baseline and STL floors fall with B ; STL stays below its paired baseline at every B .

the diagonal structure keeps the iterates diagonal, so the d -dependence reported here is a controlled best case, not a worst-case bound.

- The variance-inflation regime is reached through covariance under-dispersion; the magnitude of the benefit (and of the inflation) depends on how far C is from $C_\star$ and on the degree of target misspecification, both of which are problem-specific.

References

- [1] G. Roeder, Y. Wu, and D. Duvenaud. *Sticking the Landing: Simple, Lower-Variance Gradient Estimators for Variational Inference*. Advances in Neural Information Processing Systems, 2017.

target	scheme	B	base floor	STL floor	reduction
Gaussian	riemannian	1	0.19	$< 10^{-12}$	$> 10^{11} \times$
		4	0.05	$< 10^{-12}$	$> 10^{10} \times$
		16	0.012	$< 10^{-12}$	$> 10^{10} \times$
	kl	1	0.19	$< 10^{-12}$	$> 10^{11} \times$
		4	0.05	$< 10^{-12}$	$> 10^{10} \times$
		16	0.012	$< 10^{-12}$	$> 10^{10} \times$
log-cosh	riemannian	1	0.19	1.9×10^{-4}	$1.0 \times 10^3 \times$
		4	0.048	4.8×10^{-5}	$1.0 \times 10^3 \times$
		16	0.012	1.3×10^{-5}	$9.7 \times 10^2 \times$
	kl	1	0.19	1.8×10^{-4}	$1.1 \times 10^3 \times$
		4	0.048	4.6×10^{-5}	$1.1 \times 10^3 \times$
		16	0.012	1.1×10^{-5}	$1.1 \times 10^3 \times$

Table 2: Tail noise floor (median over configs and seeds), baseline vs. STL, with the STL reduction factor, by target, covariance scheme, and mini-batch size B .

quantity	value
backend / device	torch / NVIDIA H200 NVL
GPU workers (cap 2)	2
estimator wall time	13.1 s
algorithm wall time	1135.3 s
algorithm cells	252
mean Riemannian cell	7.69 s
mean KL cell	8.85 s

Table 3: Runtime and environment. Mean wall time is reported per algorithm cell (all seeds of one config/method/batch advanced in a single batched pass).